



Chapter 6

Replacement Analysis

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6 Replacement Analysis

(5 hours)

- 6.1 Replacement strategies (Asset life and selection of challengers over defenders)
- 6.2 Economic service life of an asset
- 6.3 Replacement strategy for asset (Project with finite and infinite planning horizon)

Definition

Replacement analysis is a financial and engineering technique used to determine whether an existing asset (the defender) should be kept in service or replaced by a new, more efficient alternative (the challenger). The primary goal is to identify the optimal time for replacement to minimize total costs or maximize profit.

Key Components

- **Defender:** The existing equipment currently in use.
- **Challenger:** The best available new alternative being considered for replacement.
- **Economic Life:** The time period that results in the lowest Equivalent Uniform Annual Cost (EUAC) for owning and operating an asset.
- **Sunk Cost:** Past expenses (like the original purchase price) that cannot be recovered. In a rational replacement study, only current market value and future costs are considered; sunk costs are ignored

Definition

Reasons for Replacement

- Physical Deterioration: Wear and tear leads to rising maintenance costs, decreased efficiency, and more frequent breakdowns.
- Obsolescence: Technological advances make newer models much cheaper or more productive to operate, even if the old asset still functions.
- Inadequacy: The existing equipment can no longer meet increased production demands or capacity requirements.

Definition

Types of Replacement Problems

- Items that Deteriorate: Assets like trucks or machinery where maintenance costs increase gradually over time.
- Sudden Failure Items: Low-cost components like light bulbs or transistors that work perfectly until they fail completely.
- Individual Replacement: Replacing each unit immediately upon failure.
- Group Replacement: Replacing all units in a system at fixed intervals to reduce labor costs, even if some are still working.

Definition

Methods of Analysis

1. **Annual Cost Method:** Compares the EUAC of the defender against the minimum EUAC of the challenger.
2. **Present Worth Analysis:** Evaluates the total present-day cost of keeping the defender versus buying the challenger over a specific study period.
3. **Marginal Cost Analysis:** A year-by-year check to see if the cost of keeping the defender for one more year is higher than the average annual cost of the challenger.
4. **MAPI Method:** A specialized method used to simplify equipment replacement decisions by considering the "adverse minimum" values.

Replacement strategies (Asset life and selection of challengers over defenders)

Replacement strategies involve an economic comparison between an existing asset (defender) and a potential replacement (challenger) to identify the optimal time to switch and minimize total costs.

Asset Life Concepts

- Selection is based on comparing the **Economic Service Life (ESL)** of both assets rather than their physical life.
- **Physical Life:** The total duration an asset can function before it is no longer usable.
- **Economic Service Life (ESL):** The period that results in the minimum Equivalent Uniform Annual Cost (EUAC).

ESL is typically shorter than physical life because operating and maintenance costs eventually rise high enough to outweigh the benefits of keeping the asset.

The goal is to find the year where the sum of annual capital recovery and operating costs is at its lowest point

Replacement strategies (Asset life and selection of challengers over defenders)

Decision Rules for Selection

The primary decision rule is based on comparing the marginal cost of the defender against the minimum EUAC or AEC (Annual equivalent cashflow) of the challenger:

Scenario	Strategy / Action
Defender Marginal Cost < Challenger	Retain the defender for at least one more year.
Defender Marginal Cost > Challenger	Replace the defender with the challenger immediately.
Defender EUAC < Challenger EUAC	Keep the defender until its cost for one more year exceeds the challenger's average annual cost.

Replacement strategies (Asset life and selection of challengers over defenders)

Key Strategies and Assumptions

- Outsider Viewpoint (Opportunity Cost): The defender's "initial cost" is its current market value (what you lose by not selling it now), while the challenger's cost is its full purchase price.
- Ignore **Sunk Costs**: **Money already spent on the defender** (original purchase price or past repairs) is irrelevant to the replacement decision and should be ignored.
- Repeatability: Often assumes the current best challenger will continue to be available at the same cost in future cycles.
- Cash Flow vs. Opportunity Cost: The Opportunity Cost approach is more common, treating the proceeds from selling the old machine as the "investment" required to keep it

Replacement strategy for asset (Project with finite and infinite planning horizon)

Replacement analysis under a finite planning horizon is the evaluation of replacement alternatives over a fixed and specified study period (N years), where all feasible replacement strategies must terminate at the end of the planning horizon.

A finite horizon means:

The project has a fixed planning period (e.g., 5 years, 8 years, 10 years).

After that period, the project ends.

Objectives

- To compare feasible replacement alternatives within a fixed study period.
- To determine the most economical time to replace an existing asset.
- To account for operating, maintenance, salvage, and capital costs.
- To incorporate the opportunity cost of retaining the defender.
- To ensure all alternatives are evaluated on an equivalent economic basis.
- To determine the optimal replacement policy that minimizes the total present worth of cost (or maximizes net present worth) over the fixed planning horizon.

Principles

- Sunk cost is ignored.
- Current market value of defender is treated as opportunity cost.
- All alternatives must be evaluated over the same planning horizon.
- Present worth method is used for comparison.

Replacement strategy for asset (Project with finite and infinite planning horizon)

Replacement analysis under an infinite horizon is the process of determining the optimal economic life of an asset when it is assumed that the service will be needed indefinitely and identical replacement cycles will continue forever.

The infinite horizon approach assumes:

- The asset will be replaced by identical assets.
- The same cost pattern repeats.
- The service requirement continues forever.

Replacement strategy for asset (Project with finite and infinite planning horizon)

Objectives

- To determine the economic service life that minimizes the Equivalent Uniform Annual Cost (EUAC).
- To find the optimal replacement interval.
- To minimize long-term average annual cost.
- To determine how long an asset should be kept before replacement.
- To compare alternatives assuming perpetual service.

Decision Criterion

Select the alternative with minimum EUAC

In infinite horizon problems:

- EUAC method is preferred.
- Present Worth is not commonly used unless converted to annual worth.

Replacement strategy for asset (Project with finite and infinite planning horizon)

Basis of Comparison	Finite Horizon Replacement	Infinite Horizon Replacement
Definition	Replacement analysis performed over a fixed and limited study period (N years).	Replacement analysis performed assuming the service is required indefinitely.
Planning Period	Fixed (e.g., 5 years, 8 years, 10 years).	Unlimited / continues forever.
End Condition	Project or activity ends after N years.	No defined end; service continues permanently.
Main Objective	Minimize total Present Worth (PW) of cost over N years.	Minimize long-term Equivalent Uniform Annual Cost (EUAC).
Method Used	Present Worth (PW) or Net Present Worth (NPW).	EUAC (Equivalent Uniform Annual Cost).
Replacement Pattern	Replacement must fit within study period.	Replacement cycles repeat indefinitely.
Salvage Value Treatment	Considered at end of planning horizon.	Considered at end of each replacement cycle.
Decision Criterion	Select alternative with minimum PW (cost problem).	Select alternative with minimum EUAC.
Sunk Cost Treatment	Ignored.	Ignored.
Typical Application	Temporary projects, contracts, limited-duration operations.	Ongoing business operations, public utilities, permanent services.

Replacement strategy for asset (Project with finite and infinite planning horizon)

Finite Horizon Examples in Civil Engineering Field

- Study period is fixed (project/contract based)
- Construction equipment purchased for a 5-year highway project.
- Concrete batching plant installed for 4-year hydropower construction phase.
- Excavator used for 6-year metro rail construction.

Examples of Infinite Horizon

- A cement factory operates indefinitely and replaces kilns periodically.
- Municipal water supply pumps operate permanently.
- Highway maintenance equipment used continuously by government.

Questions

Determine whether the existing asset (Defender) should be retained or replaced by a new asset (Challenger) under the following conditions:

- The defender has a current market value of Rs. 5,000.
- A challenger can be purchased for Rs. 7,500.
- Both assets have a service life of 3 years.
- There is no salvage value at the end of 3 years.
- The MARR (Minimum Attractive Rate of Return) is 12%.

Year	Defender (Rs.)	Challenger (Rs.)
1	1,700	500
2	2,000	1,100
3	2,500	1,300

Questions

Given Data

Defender (D) current market value = Rs. 5,000

Challenger (C) purchase cost = Rs. 7,500

Life of both = 3 years

Salvage value = 0

MARR = 12%

$$(P/F, i, n) = \frac{1}{(1 + i)^n}$$

- $(P/F, 12\%, 1) = 0.8929$,
- $(P/F, 12\%, 2) = 0.7972$,
- $(P/F, 12\%, 3) = 0.7118$

$$(A/P, i, n) = \frac{i(1 + i)^n}{(1 + i)^n - 1}$$

- $(A/P, 12\%, 3) = 0.41635$

Questions

Compute AEC of Defender (D)

Present Worth of Operating Costs (D)

$$PW_{OC,D} = 1700(0.8929) + 2000(0.7972) + 2500(0.7118)$$

- Year 1: $1700 \times 0.8929 = 1,518$
- Year 2: $2000 \times 0.7972 = 1,594$
- Year 3: $2500 \times 0.7118 = 1,780$

$$PW_{OC,D} = 1518 + 1594 + 1780 = 4892$$

Questions

Total Present Worth (D)

$$PW_D = 5000 + 4892 = 9892$$

Annual Equivalent Cost (D)

$$AEC_D = PW_D(A/P, 12\%, 3) = 9892(0.41635) = 4119$$

Questions

Compute AEC of Challenger (C)

Present Worth of Operating Costs (C)

$$PW_{OC,C} = 500(0.8929) + 1100(0.7972) + 1300(0.7118)$$

- Year 1: $500 \times 0.8929 = 446$
- Year 2: $1100 \times 0.7972 = 877$
- Year 3: $1300 \times 0.7118 = 925$

$$PW_{OC,C} = 446 + 877 + 925 = 2248 \approx 2249$$

Total Present Worth (C)

$$PW_C = 7500 + 2249 = 9749$$

Questions

Annual Equivalent Cost (C)

$$AEC_C = PW_C(A/P, 12\%, 3) = 9749(0.41635) = 4059$$

AEC(C) = Rs. 4,059 per year

$$4119 - 4059 = 60$$

Challenger saves Rs. 60 per year.

Since the Annual Equivalent Cost (AEC) of the challenger (Rs. 4,059) is less than that of the defender (Rs. 4,119), the challenger is economically preferable. Therefore, the defender should be replaced immediately.

Decision Framework

Decision Framework

The decision framework is a structured way to plan asset replacements over time. It shows which asset to use and for how many years within the planning horizon.

- The notation (J, n) means asset type J is used for n years. It clearly represents the replacement sequence step by step.
- In the example $(J_0, 2), (J_1, 5), (J_2, 3)$, the defender is kept for 2 years. Then it is replaced by asset type 1 for 5 years and later by asset type 2 for 3 years.
- The total planning horizon is the sum of all years in the sequence. In this example, $2 + 5 + 3$ equals 10 years.
- The economic life of an asset does not always mean it should be replaced at that time. The best replacement time depends on challenger costs and planning horizon as well.
- In finite horizon problems, multiple replacement sequences may be possible. We compare all feasible sequences using the Present Worth method.

Decision Framework

- In infinite horizon problems, the replacement cycle repeats forever. The goal is to find the cycle length that minimizes annual cost (EUAC).
- The main idea of this framework is to organize replacement decisions logically. It ensures we select the most economical sequence over the entire planning period.
- Under an infinite planning horizon, service is required for a very long or indefinite period. The asset will either continue in use or be replaced repeatedly over time.
- The first step is to compute the economic life of both the defender and the challenger. We also calculate their Annual Equivalent Cost (AEC) at their respective economic lives.
- The economic life is the number of years that minimizes the annual equivalent cost. It represents the most economical duration to keep an asset.

Decision Framework

- After calculating AEC of defender AEC (D) and challenger AEC(C), we compare them. This comparison tells us whether replacement is justified now.
- If AEC (D) is greater than AEC(C), it is more expensive to keep the defender. Therefore, the defender should be replaced immediately.
- If AEC (D) is less than AEC(C) it is cheaper to keep the defender. In that case, the defender should not be replaced now.

Questions

A firm has a contract to provide printing services to IOE for the next 8 years. The firm can provide the service either by continuing to use its old printing machine (current defender) or by purchasing a new printing machine (challenger).

After the 8-year contract period, neither the old machine nor the new machine will be retained.

The Annual Equivalent Costs (AEC) of the old and new machines at a MARR of 10% are given below:

Life N (years)	Old machine AEC (Rs)	New machine AEC (Rs)
1	5,15,000	7,50,000
2	5,10,000	6,15,000
3	5,50,000	5,86,000
4	5,96,000	5,83,000
5	6,44,000	5,90,000

1. Determine the economic service life of the old machine and the new machine.

2. Determine the best replacement strategy for the 8-year contract period.

Questions

Contract period = 8 years

MARR = 10%

Annual Equivalent Cost (AEC):

Old machine - Minimum AEC = Rs. 5,10,000 at 2 years ;

New machine - Minimum AEC = Rs. 5,83,000 at 4 years

Economic Service life of old machine = 2 years

Economic Service life of new machine = 4 years

Possible Replacement Strategies (Total = 8 years)

- Replace now $\rightarrow (0,4,4)$

- Keep old 1 year $\rightarrow (1,4,3)$

- Keep old 2 years $\rightarrow (2,4,2)$

- Keep old 3 years $\rightarrow (3,4,1)$

- Keep old 4 years $\rightarrow (4,4)$

$(J_0, 0)$: Do not keep the old machine at all $\rightarrow$ replace immediately (now).

$(J, 4)$: Buy the new machine and use it for 4 years.

$(J, 4)$: After 4 years, replace it again with the same new machine and use for another 4 years.

Questions

Strategy	Sequence
1	$(0,0)(1,4)(1,4)$
2	$(0,1)(1,4)(1,3)$
3	$(0,2)(1,4)(1,2)$
4	$(0,3)(1,4)(1,1)$
5	$(0,4)(1,4)$

Questions

Each strategy is divided into blocks like:

- Old machine for some years
- New machine for some years

For each block:

$$PW = AEC \times (P/A, i, n)$$

If that block starts later (not at year 0), we further discount it:

$$PW@0 = [AEC(P/A, i, n)](P/F, i, t)$$

Where

n = duration of block

t = starting year of block



Questions

$$(P/A, 10\%, 8) = 5.3349$$

$$PW_{(10\%)1} = 5,83,000 \times (P/A, 10\%, 8)$$

$$PW_{(10\%)1} = 5,83,000 \times 5.3349$$

$$PW_{(10\%)1} = 31,10,247$$

Strategy-I means replace immediately and use new machine in two 4-year cycles; its total present worth cost is Rs. 31,10,247.

Questions

Keep old for 2 years, then new for 4 years, then new for 2 years.

Total = 8 years.

Old machine for 2 years (starts at year 0)

$$\text{AEC} = 5,10,000$$

$$\text{PW} = 5,10,000(\text{P/A}, 10\%, 2)$$

$$\text{PW} = 5,10,000(1.7355)$$

$$\text{PW} = 8,85,105$$

New machine for 4 years (starts at year 2)

$$\text{AEC} = 5,83,000$$

First find PW at year 2:

$$\text{PW} = 5,83,000(\text{P/A}, 10\%, 4)$$

$$\text{PW} = 5,83,000(3.1699)$$

$$\text{PW} = 18,47,452$$

Now discount back 2 years:

$$\text{PW} = 18,47,452(\text{P/F}, 10\%, 2)$$

$$\text{PW} = 18,47,452(0.8264)$$

$$\text{PW} = 15,27,230$$

Questions

New machine for 2 years (starts at year 6)

$$AEC = 6,15,000$$

First find PW at year 6:

$$PW = 6,15,000(P/A, 10\%, 2)$$

$$PW = 6,15,000(1.7355)$$

$$PW = 10,67,333$$

Now discount 6 years:

$$PW = 10,67,333(0.5645)$$

$$PW = 6,02,509$$

Total Present Worth

$$PW = 8,85,105 + 15,27,230 + 6,02,509$$

$$PW = 30,14,844$$

Questions

$$PW_4 = 5,50,000(P/A, 10\%, 3) + 5,90,000(P/A, 10\%, 5)(P/F, 10\%, 3)$$

$$(P/A, 10\%, 3) = 2.4869$$

$$(P/A, 10\%, 5) = 3.7908, (P/F, 10\%, 3) = 0.7513$$

$$= 5,50,000(2.4869) = 13,67,795$$

$$+ 5,90,000(3.7908)(0.7513) = 16,80,336$$

$$PW(10\%)_4 = 30,48,131$$

Questions

Best Replacement Strategy

Minimum PW = Rs. 30,14,844

Strategy-3 is best because it has the lowest PW.

Best sequence: (J,0,2)(J,4)(J,2)

Keep old machine for 2 years,

Then use new machine for 4 years,

Then replace with new machine for 2 years.

The strategy with the lowest Present Worth (PW) is the most economical because it represents the smallest total cost in today's value.

Therefore, in cost comparison problems, the alternative with the minimum PW is always selected.

Questions

Strategy	Sequence	PW (Rs.)
1	(J0,0)(J,4)(J,4)	31,10,247
2	(J0,1)(J,4)(J,3)	30,53,102.5
3	(J0,2)(J,4)(J,2)	30,14,844
4	(J0,3)(J,4)(J,1)	31,06,149
5	(J0,4)(J,4)	31,51,479

The strategy with the lowest Present Worth (PW) is the most economical because it represents the smallest total cost in today's value.

Questions

- Suppose a company plans to purchase a new machine with an initial cost of Rs. 18,000. The machine is expected to have a useful life of eight years and an operating cost of Rs. 3,000 in the first year. For the remaining years, the operating cost increases annually by 15% over the previous year's operating cost. The salvage value decreases each year by 20% from the previous year's salvage value. Determine the economic service life of the machine if the MARR is 12% per year

Initial Cost (P) = Rs. 18,000

Life = 8 years

Operating Cost (Year 1) = Rs. 3,000

Operating cost increases = 15% every year

Salvage value decreases = 20% every year

MARR (interest rate) = 12%

Questions

We use:

$$AEC = CR + OC$$

Where:

- CR = Capital Recovery
- OC = Operating Cost (annual equivalent)
- Choose N where AEC is minimum

Interest = 12%

$$CR = P(A/P, i, N) - S(A/F, i, N)$$

$$OC = \left[\sum OC_n(P/F, i, n) \right] (A/P, i, N)$$

Questions

For how many years should we keep the machine so that cost per year is minimum?

Operating Cost Each Year

Increase 15% every year:

Year	Operating Cost (Rs.)
1	3000
2	3450
3	3968
4	4563
5	5247
6	6034
7	6939
8	7980

Questions

Salvage Value decreases = 20% every year

Year	Salvage
1	14,400
2	11,520
3	9,216
4	7,373
5	5,898
6	4,718
7	3,774
8	3,019

Questions

$$(A/P, 12\%, 1) = 1.12$$

$$(A/F, 12\%, 1) = 1$$

$$CR = 18000(1.12) - 14400(1)$$

$$CR = 5760$$

$$OC = 3000$$

$$AEC_1 = 5760 + 3000$$

$$AEC_1 = \mathbf{8760}$$

Questions

$$(A/P, 12\%, 2) = 0.5917$$

$$(A/F, 12\%, 2) = 0.4717$$

$$CR = 18000(0.5917) - 11520(0.4717)$$

$$CR \approx 5217$$

PW of OC:

$$= 3000(0.8929) + 3450(0.7972)$$

$$= 5428$$

$$OC = 5428(0.5917)$$

$$OC \approx 3212$$

$$AEC_2 = 5217 + 3212$$

$$AEC_2 = \mathbf{8429}$$

Questions

$$(A/P, 12\%, 3) = 0.4163$$

$$(A/F, 12\%, 3) = 0.2963$$

$$CR \approx 4763$$

$$PW \text{ of OC} \approx 8253$$

$$OC = 8253(0.4163)$$

$$OC \approx 3436$$

$$AEC_3 = 4763 + 3436$$

$$AEC_3 = \mathbf{8199}$$

N	AEC (Rs.)
1	8760
2	8429
3	8199
4	8055
5	7985
6	7977
7	8024
8	8118

Continue AEC calculation for year 4,5,6,7,8 years as well.

Questions

Before 6 years → Capital cost burden reducing

After 6 years → Operating cost rising too much

So total annual cost becomes minimum at Year 6

Minimum AEC = 7977

Occurs at $N = 6$ years

Questions

Q)Nepal Telecom plans to purchase a router with an initial cost of \$18,000.The router will have a salvage value of \$12,000 at the end of the first year, which decreases by 10% each year thereafter for the remaining useful life. The annual operation and maintenance (O&M) cost is \$2,000 in the first year, and it increases by \$1,500 every year.The useful life of the router is 8 years. Determine the economic service life of the router.

Q)Explain the decision framework used in replacement analysis. Using the data provided for the defender and challenger, justify the appropriate replacement guideline when the project life is 8 years and the Minimum Attractive Rate of Return (MARR) is 15% per year.

The Annual Equivalent Cost (AEC) at 15% for the defender and challenger is given below:

N	Defender AEC (\$)	Challenger AEC (\$)
1	5,380	7,700
2	5,203	6,184
3	5,468	5,756
4	5,845	5,625
5	6,258	5,631
6	6,682	5,721

Thank you for your attention!